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BA HISTORY HONOURS

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History of Keralam Upto the 12th Century

Module I

Keralam : Landscape and Environment

MODULE-I

Keralam : Landscape and Environment

I.1. Keralam as a Region

Kerala is a narrow strip of land with an area of 38,863 square kilometres. It lies between 8°17'30" (eight degrees seventeen minutes thirty seconds) and 12°27'40" north latitude, and 74°51'57" and 77°24'47" east longitude. The Western Ghats, with their high mountain peaks, form the eastern boundary of the State, while the Arabian Sea forms the western boundary. These natural features give Kerala its unique physical character. Kerala shares its northern and northeastern borders with the Dakshin Kannada, Kodagu, and Mysore districts of Karnataka. Its eastern and southeastern borders are shared with the Tamil Nadu districts of Nilgiri, Coimbatore, Madurai, Ramanathapuram, Tirunelveli, and Kanyakumari. The shape of Kerala is like a scalene triangle, with a long coastline of about 560 kilometres forming its base and the Western Ghats forming its pointed end. The width of the State varies from about 11 kilometres at its narrowest point to about 124 kilometres at its widest point.

I.1. A. Geographical Features

Based on a detailed study of the physiography and slope of Kerala, the State can be divided into five main physical regions. These are: mountain peaks above 1,800 metres, highlands between 600 and 1,800 metres, midlands between 300 and 600 metres, lowlands between 10 and 300 metres, and coastal plains and lagoons below 10 metres. The mountains and peaks above 1,800 metres, located within the Western Ghats, make up only about 0.64% of Kerala's total area. In the northern part of the Western Ghats within Kerala, only a few peaks rise above 1,800 metres.

Highlands:
The highlands form an important physical region and cover about 20.35% of the State's area. From the far north of Kerala, these mountain ranges run parallel to the coastline up to Vavalmalai, which lies to the east of Kozhikode.

Midlands:
The midland region is made up of the gently rolling western edge of the highlands, laterite-covered rocky spurs that extend westward, and parts of the natural breaks in the earth's crust (passes). It covers about 8.44% of the total area of Kerala.

The midlands form most of the eastern part of Kannur district. However, their width decreases toward the western side of the Wayanad plateau, where they appear only as a narrow belt along the steep slopes. From the western side of the Nelliampathi plateau to the northern part of the Cardamom Hills, long rocky spurs separated by deep valleys gradually merge with the gentler slopes of the lowland region.

Lowlands:
The lowlands include areas lying between 10 and 300 metres above sea level. These areas are mainly made up of gently worn landforms known as dissected peneplains. This zone is uneven in height and covers the largest share of the State's area, about 54.17%. The lowlands contain many floodplains, alluvial terraces, filled valleys, colluvial deposits, and other sedimentary formations. In the northern and southern ends of Kerala, this region gradually merges with the coastal plains, but with steeper slopes than those seen in most other parts of the State. The higher parts of the lowlands are more common in the eastern areas of both the northern and southern districts. In places such as

Poovar–Vizhinjam, Kadakkavur–Varkala, north of Badagara, and Bekal–Kasaragod, the lowland region directly meets the sea, with no coastal plains in between.

Coastal plains and lagoons:
The broad, low-lying area along the coast is an important physical region of Kerala. It is also very important for economic activities and population distribution. This region covers about 16.40% of the total area of the State.

In central Kerala, most of the coastal plain lies about 4 to 6 metres above mean sea level. In northern and southern Kerala, the elevation is slightly higher, ranging from 4 to 10 metres, except in areas with coastal cliffs, headlands, and sloping rock platforms. This region includes beach dunes, old beach ridges, barrier flats, coastal alluvial plains, floodplains, river terraces, marshes, and lagoons. The coastal plain is widest in the Alappuzha and Aluva–Kaladi areas.

A special feature of this region is the presence of many beach dune ridges that run parallel or nearly parallel to the coastline, especially in the Alappuzha–Cherthala area. Their alignment suggests that the shorelines formed during at least two different periods. The maximum distance between the oldest and the youngest ridges, near the present shoreline, is about 18 kilometres. In addition, the coastal plain extends between several rocky ridges along the coast.

I.1.B. The Western Ghats

The Western Ghats are the most important landform of Kerala and one of the major mountain systems of Peninsular India. In ancient Indian literature, the Western Ghats were known as Sahyadri, meaning the great escarpment of the Indian subcontinent. This mountain range plays a key role in controlling the climate of South India. Most of the heavy rainfall during the south-west monsoon falls on the western slopes of the Western Ghats. The Western Ghats are made up of different types of metamorphic rocks. Granulites are found in most parts of the State, while schists occur mainly in the Wayanad and Attapadi regions. These rocks differ in composition, structure, and age.

North of the Palghat Gap, the Western Ghats generally have gentle slopes on the eastern side, while the western slopes are much steeper and often very steep cliffs. This pattern is not limited to northern Kerala alone. Similar slope features are also seen in the Western Ghats of Maharashtra and Karnataka, where, over long stretches, the hills are carved out of flat-lying Deccan Trap rocks. South of the Palghat Gap, the Western Ghats show steep escarpments on the eastern side, and the major interior plateaus slope towards the west. This is different from the landscape north of the Palghat Gap. The present landscape of the Western Ghats was shaped by the combined action of internal forces such as tectonic movements and external forces like weathering and erosion, all influenced by the type and structure of the rocks.

Evidence for the uplift of specific sections of the Western Ghats comes from detailed geological studies in areas such as Wayanad, the northern side of the Palghat Gap, the northern side of the Ariankavu Pass, and the Ponmudi block. Additional evidence comes from matching ancient erosion surfaces with known breaks in the sedimentary record of the west coast basin. Within Kerala, the Western Ghats gradually merge with the Mysore Plateau and the interior plains of Tamil Nadu. The fact that the interior plateaus slope in different directions north and south of the Palghat Gap suggests uneven tilting of the Western Ghats due to tectonic forces. The Palghat Gap itself is a major geological fracture and marks an important tectonic and rock boundary in the Kerala region.

I.1.B.i. Interior Plateaus

The Western Ghats are marked by the presence of several plateaus, which are large, mostly flat areas found at high elevations and usually associated with rugged terrain. In Kerala, the most important of these are the Wayanad Plateau, the Nelliampathi Plateau, the Munnar Plateau, and the Greater Periyar Plateau. All these plateaus occur at different height levels and show remnants of laterite caps. It is important to note that extensive laterite formation in Kerala is generally limited to areas below 600 metres in elevation. This supports the view that these plateaus were originally old, nearly level erosion surfaces. Over time, they were modified by erosion and later uplifted to different heights due to tectonic movements during various geological periods.

I.1.B.ii. Wayanad Plateau

East of longitude 75°55′ E, near Kozhikode, the Western Ghats give way to the Wayanad Plateau, which lies at an elevation of about 700 to 950 metres above mean sea level. This plateau generally slopes towards the east and northeast and gradually merges with the Mysore Plateau, which lies between 900 and 1,200 metres above mean sea level. Together, they form a large and continuous land system of Peninsular India. The Wayanad Plateau is drained by the Kabani and Moyar river systems. The change in height from the Western Ghats to the plateau is more clearly seen in the southern part than in the north, with a height difference of about 550 to 800 metres. Within the plateau, prominent isolated hills known as inselbergs and dome-shaped rock features called exfoliation domes or bornhardts are commonly seen. These landforms are usually composed of granite gneiss. Banasuramala is the highest peak in this region.

I.1.B.iii. Nelliampathi Plateau

Just south of the Palghat Gap, the Western Ghats give way to the Nelliampathi Plateau, which lies at an elevation of about 500 to 700 metres. The plateau slopes towards the southwest and is more extensive on its eastern side. Its surface is dotted with oval-shaped small hills.

The southern extension of the plateau includes the Parambikulam and Sholaiyar reservoirs. The western edge of the plateau gradually slopes down to the midland region of Thrissur district through a broad foothill zone, which is drained by the Puzhackal and Karuvannur rivers.

The northern edge of the plateau appears to be cut off by a geological lineament that aligns with the Sholaiyar River. The plateau is mainly composed of charnockitic gneiss, hornblende gneiss, and granite.

I.1.B.iv. Munnar Plateau

South of the Palghat Gap, the Western Ghats expand into a broad hill mass in the area south of the Anaimalai Hills. With curved extensions reaching into Tamil Nadu, this region forms the Munnar Plateau, which lies at an elevation of about 1,500 to 2,000 metres. It is the highest plateau in Kerala and shows remnants of laterite caps. The Munnar Plateau is bounded by the Kannandevan Hills to the north and the Cardamom Hills to the south. It is widest in the central part and appears to be part of a large, old erosion surface that slopes towards the southwest and descends in a step-like pattern.

The plateau is mainly made up of granitic gneiss, granite domes, and patches of metamorphosed sedimentary rocks. Broad U-shaped valleys and wide ridges or rounded hillocks are common features of this plateau. The southern edge of the plateau shows a sudden drop in height and coincides with a geological lineament marked by the Bodinayakkannur Pass.

I.1.B.v. Greater Periyar Plateau

South of the Munnar Plateau, the Western Ghats gradually decrease in height in the Cardamom Hills region and give way to the long and extensive Greater Periyar Plateau, which lies at an elevation of about 700 to 1,000 metres. The Greater Periyar Plateau includes two parts: the northern Periyar Plateau and the southern Pirmed Plateau. Its southern edge is sharply cut off by the Gudalur Pass.

The eastern boundary of the plateau is a straight, steep scarp that drops down to the plains of Tamil Nadu. Rivers flowing across the plateau follow unusually straight courses, and the stream network shows a clear trellis pattern. When east–west profiles are examined, steep escarpments are seen along both the western and eastern edges of the plateau. The western escarpment ends abruptly and the plateau then merges into the gently rolling plains of Kottayam district. It is made up mainly of metamorphosed sedimentary rocks, charnockitic gneiss, hornblende (biotite) gneiss, and granitic gneiss. In many areas, there is clear evidence of strong rock mixing and granite formation processes.

The Western Ghats as a Natural Barrier and Corridor in Kerala ?

The Western Ghats on the eastern side of Kerala is almost continuous and stand like a natural fort from south to north sloping to the east. It has an height of average five thousand feet from the sea level. The Western Ghats from Calicut to the Vavul mountains in the east lies parallel to the sea for a length of forty Kilometers. From this point it turns to the north-east, covers Nilambur and then goes to Palakkad. Covering the Palakkad Pass for about forty one Kilometers, the Western Ghats again heightens up and reaches to Anamala through Nelliampathy and Pothundy. The highest peak in the Ghats Anamudi stands at Anamala. From Anamala it moves through Sabarimala, Ponmudi, Agastyamudi and Mahendramudi and finally reach at Aruvamozhi. Apart from the Western Ghats, Kerala has some other minor Ghats also. The most important one among them is the Ezhimala in the northern side with a height of about two Hundred and twenty-five metres. Ezhimala situates to the north of Kannur projecting into the sea and had acted as a point of direction for the ships in the ancient times.

The Western Ghats acted like a natural barrier for Kerala from the ancient times and resisted external attacks to a certain extend. The conquerors from the other parts of India could not attack Kerala upto the 18th century because of the mountain barriers. The few who had attacked Kerala prior to the 18th century was through the Passes of Palakkad and Aruvamozhi. Some of the historians firmly believe that the Portugese who had reached Kerala in the 15th century could not spread to the other parts of India due to the Western Ghats.

I.1.C. Passes

The Western Ghats and the surrounding rugged highlands of Kerala extend continuously from north to south, except at a few breaks. The most important of these breaks is in the Palghat region and is known as the Palghat Gap. Other smaller breaks occur near Bodinayakkannur, Gudalur, and Ariankavu. The Palghat Gap is the only major break in the Western Ghats. It connects the west coast of India with the peninsular region to the east and is a clearly defined landform in South India. The gap runs in an east–west direction for about 80 kilometres and has a width of roughly 30 kilometres. Although the Palghat Gap itself is a low and gently shaped area, with an average height of about 60 metres in the west and 120 metres in the east, it is bordered by very high mountains. The Nilgiri Hills to the north and the Anaimalai Hills to the south surround the Gap, and many peaks in these ranges rise above 2,000 metres.

I.1.C.i. The Palakkad Pass played a crucial role in historical interactions:

It was through the Palakkadan Pass that the Cheras of Kerala extended their territories to Kongu Nadu in the ancient times and the Mysoreans attacked Kerala in the 18th century. Hyder Ali, the Mysore Sultan understood the strategic importance of the Palakkadan Pass and constructed a Fort in 1776. The Kerala-Tamil Nadu relations developed through the Palakkadan Pass.

I.1.C.ii. Other Passes

The other passes in the Western Ghats are at higher elevations but are narrower than the Palghat Gap. For example, the Gudalur Pass lies at about 900 metres and the Ariankavu Pass at about 400 metres. The Bodinayakkannur and Gudalur passes are located in the High Ranges, within the Cardamom Hills, and provide road access to the plains of Tamil Nadu. The Bodinayakkannur Pass aligns with east–west flowing smaller streams, while the Suruli River flows through the northeast–southwest trending Gudalur Pass.

The southernmost pass in Kerala is the Ariankavu Pass, located in the Punalur region. The pass is about 2.5 km wide, with an elevation of roughly 400 metres, while the surrounding flanks rise above 800 metres. The rocks found in the pass include strongly sheared khondalites, migmatitic gneisses, and patches of charnockite. The occurrence of small earthquakes along these passes supports the idea that they are tectonic in origin. For example, tremors were recorded in the Idukki–Kambam region on 7th and 8th June 1988 along the Bodinayakkannur Pass, and in the Punalur region along the Ariankavu Pass on 17th September 1993 and 14th November 1993.
Recent geothermobarometric studies of rocks from the northern flank of the Ariankavu Pass show a cooling history consistent with crustal uplift. However, the amount of uplift here is smaller, around 2 km, compared to the Palghat Gap region, where the uplift is estimated to be 5–6 km.

Other Important Routes

The other Passes in the Western Ghats apart from the Palakkadan Pass, are the Cherambadi Pass which connects Kerala with Coorg, the Periya and Tamarassery Passes which connect Kerala with Mysore through Wynad. The Bodinaikannur route connects Idukki with Madurai and the Aryankavu route connects Quilon with Tirunalveli in Tamil Nadu.

I.1.D. Indian Ocean

The Arabian Sea is the next important geographical division of Kerala to influence its historical evolution. The long coastal line has provided a naval tradition for Kerala of its own and many natural ports. The sea coast of Kerala could attract visitors and merchants from Europe and the other parts of Asia from the ancient period itself. The ancient ports like Muziris, Tindis, Barecca, Nelkinda etc. and the later ports like Quilon, Kochi, Kozhikode etc. have helped Kerala to develop its own history through the ages. The foreigners, especially the Europeans called this part of the world Malibar or Malabar because of its geographical peculiarity. The principal religions of the world arrived Kerala through the sea. The Jews, Christians and the Muslims reached Kerala through the Arabian sea coast.

The Kerala coast had developed trade relations with the Mediterranean world and the Persian Gulf region from the ancient period itself. It had also maintained trade relations with south-east Asia through the Bay of Bengal during the ancient times. Documentary evidence is available to prove that the emissary of the Chinese Emperor had reached the Kerala coast for pepper during the 1st century A.D. The trading activities between China and Kerala continued for many centuries. The Monsoon wind in the Arabian sea helped the Roman ships to reach the Kerala coast in the ancient period itself. Many material and literary evidences have provided informations about the existence of the brisk trade between the Roman empire and Kerala in the ancient period.

In the ancient period sea water had pierced into many places in the coastal land of Kerala. At certain places the land level was much lower to that of the sea level. So it was easy to make salt in the coastal regions in those days. The sea coast also helped for the growth of fishing.
The sea coast continued its influence upon the history of Kerala during the modern period also. It helped for the advent of the Portugese by the end of the 15th century and their subsequent domination in South India. The Dutch and the French also arrived in Kerala through the same sea

coast. The English East India Company also used the Kerala sea coast for the extension of their imperialistic domination all over India during the 18th century. The remains of the European forts at Anjengo, Tankassery, Pallippram, Thalassery and Kannur carry the memoirs of the European domination over the Kerala sea coast. More freedom at the Kerala sea coast was given to the Europeans by the ruling Naduvazhis who were interested in the foreign trade with the Europeans. When the European relations of the earlier period was limited to cultural and trade relations only, it became the political domination of the European countries in the modern period.

I.1.E. Rivers

Kerala is drained by 44 rivers, of which only three flow eastward. Most rivers originate in the Western Ghats, and they are short, fast-flowing, and display different stages of development. In their upper reaches, these rivers often form cascades and waterfalls, while in the plains they show signs of mature development. Some rivers have steep gradients of 1 in 250 or more in their upper courses. For example, the Periyar and Chalakudi rivers maintain such steep slopes for nearly three-fourths of their length. Steep gradients are also seen in the upper reaches of rivers like the Chaliyar, Valapattanam, Vamanapuram, and Karamana, indicating that these rivers are still in a youthful stage of development.

Among Kerala's rivers, the Periyar, Bharathapuzha, Chaliyar, Pamba, and Chalakudi rivers are notable for their length and the size of their drainage areas. These rivers, along with the Uppala River, flow entirely within Kerala, though some originate outside the State. The east-flowing rivers Kabani, Bhavani, and Pambar originate in Kerala but flow into Karnataka and Tamil Nadu, eventually joining the Cauvery River. Some important small streams in north Kerala include the Kumbla, Kalnad, Bekal, and Purapparamba, which either have independent outlets or flow directly into backwaters. In south Kerala, the Ulloor Thodu, which drains parts of Thiruvananthapuram city and suburbs, and the Vizhinjam Thodu, south of Kovalam, are notable streams of this category.
The five major rivers—Periyar, Bharathapuzha, Pamba, Chaliyar, and Chalakudi—together drain about 40% of the total geographical area of Kerala. Brief descriptions of these rivers are provided below.

I.1.E.i. Periyar
The Periyar River, with a length of 244 km, is the longest river in Kerala. It is also the largest in terms of drainage potential, covering an area of 5,398 km², of which 114 km² lies in Tamil Nadu. Its catchment spreads across the districts of Idukki and Ernakulam.
The river is formed by the confluence of numerous streams originating from the Sivagiri Hills in the Western Ghats at an elevation of about 1,800 metres. In its initial course, it flows northwards and passes through several gorges, the most notable being the Idukki Gorge.
Several major reservoirs are located within the Periyar basin, including Idukki, Periyar, Anairangal, Mattupatti, and Setuparvatipuram. The river flows through a metamorphic terrain.

I.1.E.ii. Bharathapuzha
The Bharathapuzha River is the second longest river in Kerala, with a length of 209 km and a catchment area of about 4,400 km² within the State. It originates in the Anaimalai Hills at an elevation of 1,963 metres and enters Kerala through parts of the Coimbatore district in Tamil Nadu. The river drains the districts of Palakkad, Malappuram, and Thrissur before emptying into the Arabian Sea. The river is formed by the confluence of four main tributaries: Gayatripuzha, Kannadipuzha (Chitturpuzha), Kalpathipuzha, and Thuthapuzha. It drains into the sea through the Ponnani estuary, which remains open throughout the year. During the monsoon season, large stretches of the river are navigable, but in the dry season, most of the riverbed dries up, except for a few kilometres near the mouth.

Major reservoirs within the Bharathapuzha basin include Mangalam, Pattambi, Chulliar, Minkara, Walayar, and Malampuzha.

I.1.E.iii. Chaliyar (Beypore River)

The Chaliyar is the third largest river in Kerala by catchment area, covering 2,923 km², after the Periyar and Bharathapuzha. It originates in the Ilambaleri Hills in Wayanad at an elevation of 2,068 metres and flows through the districts of Malappuram and Kozhikode.

Major tributaries of the Chaliyar include Chalipuzha, Punnapuzha, Pandiyar, Karimpuzha, Cherupuzha, Kanjirapuzha, Kurumanpuzha, Vadapuram Puzha, Irinjipuzha, and Iruthilly Puzha. The area around the confluence of the Karimpuzha and Punnapuzha is known for placer gold deposits. The river has a total length of 169 km and empties into the Arabian Sea west of Feroke, near Beypore.

I.1.E.iv. Pamba

The Pamba River is the third longest river in Kerala at 176 km and has the fourth largest catchment area of 2,235 km². It originates in the hill ranges of Pathanamthitta district and the adjoining Peerumed Plateau. The major reservoirs in the Pamba basin are Pamba and Kakki.

I.1.E.v. Chalakudi

The Chalakudi River is formed by the confluence of five streams: Parambikulam, Kuriaarkutty, Sholayar, Kaarapara, and Anakkayam. Among these, the Parambikulam and Sholayar rivers originate in the Anaimalai Hills at elevations of 1,733 m and 1,332 m, respectively.

Major reservoirs in the basin include Peringalkuthu, Sholayar, and Parambikulam.

I.1.F. Backwaters

A distinctive feature of the Kerala coast is the chain of water bodies called “kāyals” (backwaters), which run parallel or at an angle to the coastline. Most of these are interconnected by natural or man-made canals, allowing navigation along nearly the entire coast of Kerala. A kāyal is generally a body of brackish or hypersaline water, separated from the sea by a sandy barrier, with an inlet connecting it to the open sea. Many perennial rivers discharge into these backwaters. The southern half of Kerala has a greater number of larger backwaters.

Most of the Kerala backwaters are separated from the sea by elongated sand bars, and can therefore be classified as coastal lagoons. Since perennial rivers flow into these water bodies, they form a compound system, which can be considered a lagoonal–estuarine or partially mixed estuarine system. These water bodies may have originated as lagoons, forming in low-lying coastal areas where now-straight youthful rivers initially discharged. Changes in river courses in Kerala may be linked to the closure of old outlets or the formation of new lagoons, possibly associated with tectonic activity.

During the monsoon, the backwaters overflow into the sea, carrying large amounts of sediment, while in the summer, seawater penetrates far inland into the kayals.

Among Kerala’s estuaries, the most prominent are the Vembanad and Asthamudi kayals, which are briefly described below.

I.1.F.i. Vembanad Lake

Vembanad Lake, the largest estuary in Kerala, extends from Cochin (Kochi) to Alleppey (Alappuzha) over a distance of 83 km. Its width ranges from a few hundred metres to 15 km, and it is elongated in a NW–SE direction. Six major rivers—Periyar, Muvattupuzha, Meenachil, Manimala, Achankovil, and Pamba—discharge into the lake.

I.1.F.ii. Asthamudi Kāyal

Asthamudi Kāyal is distinctive due to its unique shape and extent, with eight branches radiating from near its central part—hence the name Ashtamudi, meaning "eight braids." It extends from north of Quilon (Kollam) town, with a total length of 16 km and a maximum width of about 15 km. The average width of each individual lobe is roughly 3 km. Its longitudinal direction runs roughly perpendicular to the coastline.

The deltaic island of Munrothuruthu, at the mouth of the Kallada River, has accreted around a lateritic outlet with aquifer strata. Multiple water bodies and paleo-river channels around the Kallada River suggest that the estuary was more extensive in the past.

According to the Kerala Public Works Department, the major lagoons in the State include: Kavvayi, Agalapuzha, Enamakkal-Manakkodi (ഏനാമാക്കൽ, മനക്കൊടി, Muriad (മുരിയാട് (തൃശൂർ), Kodungallur-Varapuzha, Sasthamkotta, and Vellayani. Among these, Sasthamkotta and Vellayani are significant sources of freshwater supply.

I.1.G. Soils of Kerala

The soils of Kerala have been classified into ten broad groups base on the morphological features and physico-chemical properties. These are (i) red loams, (ii) lateritic soil, (iii) coastal alluvial soil, (iv) riverine alluvial soil, (v) greyish Onattukara soil (vi) brown hydromorphic soil (vii) hydromorphic saline soil, (viii) acid saline soil, (ix) black soil and (x) forest loam.

I.1.G.i. Red Loams

Red loams are localised in the southern parts of Trivandrum district. The soil is highly porous, acidic in reaction, essentially kaolinitic in nature and poor in organic matter and in all essential plant nutrients.

I.1.G.ii. Lateritic Soils

Lateritic soils are the most predominant in Kerala. These are mostly reddish brown to yellowish red in colour. The texture of surface soils ranges mostly from gravelly loam to gravelly clayey loam. The morphologic characteristics vary to some extent depending on the rock type. In Kozhikode, Malappuram, Cannanore (Kannur) and Kasaragod districts extensive stretches of indurated laterites with hard surface crust are common. These are generally poor in available nitrogen, phosphorus and potash and also low in bases. The profiles are generally deep, well-drained and respond well to management practices. A wide range of crops such as coconut, tapioca, rubber, arecanut, pepper, cashew etc. are cultivated in this soil.

I.1.G.iii. Coastal Alluvial Soils

Coastal alluvial soils are seen along the west coast tract. These have developed from recent marine deposits. Sand is the prominent fraction. The horizon is usually thin and the surface textures observed are loamy sand to sandy loam. The water table is high in low-lying areas.

I.1.G.iv. Riverine Alluvial Soils

These occur mostly along banks of rivers and their tributaries. They show wide variations in their physico-chemical properties and in the arrangement of layers depending on provenance features. Although these soils show a general predominance of sand fraction, silt fraction is dominant in Quilon taluk. Presence of mica flakes has been observed in alluvial soils of Ernakulam district. Deposition of organic matter is observed in 'kole' lands of Thrissur. These soils respond well to management practices and are ideally suited for cultivation of coconut, Paddy, arecanut, pepper, tapioca etc.

I.1.G.v. Greyish-Onattukara Soils

These are confined to the region of Karunagappally, Karthikappally and Mavelikkara taluks of Quilon and Alleppey districts. They occur as marine deposits extending into the interior upto the laterite belt and consist of coarse fractions. Water table is high in these profiles. Coconut is the major crop grown with less paddy and tapioca.

I.1.G.vi. Brown Hydromorphic Soils

These occur quite extensively, occupying valley bottoms. They exhibit wide variation in physico-chemical properties and morphological features, possibly resulting from transportation and deposition under varying conditions and provenances. These are generally very deep brown in colour. Surface texture varies from sandy loam to clay. In many instances pottery-quality clay occurrences are associated with these soils. Occurrence of laterite and gravel in these soils is indicative of the colluvial factor in soil formation.

I.1.G.vii. Hydromorphic Saline Soils

These are seen in the coastal tracts of Ernakulam, Alleppey, Thrissur and Cannanore districts. These are not comparable in origin with the saline soils occurring in other parts of the country. These are generally brownish, deep and poorly drained. The profiles show wide variation in texture. Salinity here is caused by the inflow of tidal waters into these areas through the network of backwaters and estuaries. During rainy season in June-July, the areas are almost free of salts, while the maximum salinity is observed during March-April. One crop of paddy is raised as monocrop during August-December.

I.1.G.viii. Acid Saline Soils

These are confined to the Kuttanad region, a substantial portion of which lies below sea level, and is submerged under water for a major part of the year. The area is susceptible to seasonal ingress of saline water as a result of saline inflow from the sea, and during the monsoon seasons receives fresh water from the rivers. The soils, therefore, form the typical water-logged soils and are entirely different from normal well-drained soils in their morphological, physical and chemical characteristics. Three varieties, namely (1) Kāyal soils, (2) Karapadam soils and (3) Kari soil have been identified.

I.1.G.viii.a. Kāyal Soils

Kāyal soils are found in the reclaimed lake beds in Kottayam and Alleppey districts. These are very deep, poorly drained, dark brown alluvial silty-clay loams with occasional limeshells. The clay content decreases towards depth, and the soils exhibit typical aquic characters such as grey colour, red and brown mottles and streaks. These are slightly acidic, medium in organic matter and poor in total available nutrients, but are fairly rich in calcium. Accumulation of salt on the surface often leaves a white colour.

I.1.G.viii.b. The Karapadam Soils

The Karapadam soils occur along the inland water-ways and rivers and are distributed over a large part of upper Kuttanad. These are river-borne alluvial soils occurring in flat lands, 1 to 2m below sea level. These are very deep, poorly drained, dark grey clay loams and silty clay subsoils. Presence of sand pockets are observed. High acidity, high salt content and fair amounts of decayed organic matter are characteristic of the soils. Available nutrients, particularly phosphorus is generally poor, and lime deficiency is also noticed.

I.1.G.viii.c. Kari soils

Kari soils are found in patches in the districts of Alleppey, Kottayam and Ernakulam. These are black and poorly drained soils, occurring 1-2 m below sea level, and resemble peat soils and exhibit characteristics of submerged forest area. Decomposed organic matter is often observed in lower layers. The soils show typical aquic characteristics and are highly acidic in reaction.

I.1.G.ix. Black Soils
Patches of black soils are found to occur in parts of Palghat district and these are considered as extension of the black soils observed in the adjacent Coimbatore district in Tamil Nadu. These are dark in colour, low in organic matter, calcareous, moderately alkaline, high in clay content and cation exchange capacity and are very sticky and plastic. The soils occur in gently sloping to nearly level lands and crack during dry periods.

I.1.G.x. Forest Loams
These are restricted to the eastern part of the State, and are developed over the crystalline rocks of the Western Ghats. In areas with lesser canopy, signs of lateritisation predominate over soil formation, indicating the modifying effects of vegetation over soil formation. These are dark reddish-brown to black in colour with loam to silty loam in texture, and have wide variation in depth.

I.1.H. Climate

I.1.H.i. Monsoons
Kerala has a tropical climate, dominated by the monsoon. The steep variation in relief—from the western coastal plains to the hilly regions of the Western Ghats in the east—and proximity to the sea influence local climate. In the eastern highlands, some areas experience a sub-tropical climate. Gaps in the Western Ghats, such as the Palghat Gap and Ariankavu Pass, allow heat from the Tamil Nadu plains to influence local weather, particularly in Palghat and Punalur.

I.1.H.i.a. Rainfall

Kerala experiences two monsoons, namely the southwest (June to September) and the northeast (October to December) seasons. Locally these are known as "Edavappathi" and "Thulavarsham", respectively. The annual rainfall varies from less than 100 cm (around Chinnar and north of Marayur) to more than 500 cm (around Neriyamangalam) with an average of about 300 cm, precipitated mainly during southwest and northeast monsoons. Occasional rainfall is also received between the two seasons. Rainfall exceeding 50 cm during the latter period is recorded around Hosdurg, Kozhikode, north of Cochin (Kochi) and in the foothills of Periyar plateau.
The southwest monsoon, yielding more than 60% of the total precipitation, records high rainfall of more than 300 cm in the foothill zones of Wayanad plateau and Pirmed-Neriyamangalam stretch.
The northeast monsoon yields less rainfall, with the maximum (>100 cm) recorded in Neriyamangalam. The northern part, generally receives less rain (8% of annual rainfall) in this season, whereas in the south, it is about 504 mm average (24% of the annual rainfall).
Further, the maximum amount of rainfall is recorded in areas where the Western Ghats have higher elevation and farther from the sea (eg Kozhikode, Peerumed-Neriyamangalam etc). The southernmost districts of Trivandrum receive the minimum, where the Ghats are lower in elevation and are relatively closer to the coast.

I.1.H.i.b Temperature

Temperature data for the years 1931 to 1979 indicate that the period from March to May is the hottest with temperatures reaching more than 32°C and the lowest temperatures are experienced during the months of December and January. The highest temperature ever recorded at an individual station is 41.71°C at Palghat on 26th April, 1950 and the lowest temperature on record was 12.9°C at Punalur on 8th January, 1965. Temperature is minimum in the month of July, when the State receives plenty of rainfall and the sky is cloudy. It is 28.5°C for the State as a whole.
Mean annual temperatures for the period also indicate that most of the lowland and midland regions experience temperatures in the range of 25.5 to 27.5°C, while higher temperatures (>27.5°C) are experienced in a belt in central Kerala, comprising midland parts of Malappuram, Thrissur, Alleppey and Kottayam.

In the coastal belt, the temperature goes upto 30-32°C while in the Palghat Gap region often the temperature goes beyond 38°C . The lowest mean annual temperature (17.5 to 19.5°C) is experienced in the Munnar-Devikolam area of the Western Ghats. Wayanad plateau and the hilly tracts of the Western Ghats experience moderate temperatures (21.5 to 25°C). Maximum temperature rises gradually from July onwards till March. Night temperature starts rising gradually from January onwards till May, and it has a secondary minimum in July when the State receives plenty of rain. The increase in maximum temperature from July to March generally varies from 3.2 to 9.8°C.

I.1.H.i.c. Wind

Winds over the State are seasonal and notable change in wind direction and speed is discernible between the coastal and inland regions of the State, as well as between various localities. Cannanore (Kannur) and Kozhikode in north Kerala are dominated by wind from northwest.
In the Palghat Gap region, winds are predominantly from the east during November-March and from the west for the remaining period of the year. In Cochin and Alleppey, the wind is mainly from the west and northwest, while it is from northwest in Trivandrum. Generally, winds from northeast and east prevail in the morning, while in the afternoon it is from west and northwest. In a number of places, including the Palghat Gap, wind speed exceeds 15 km/hour for a couple of hours daily. Wind speed attains 40-50 km/hour during and before the monsoon rains.

I.1.H.i.d. Humidity

Annual mean relative humidity varies from 79 to 84% in the morning to 73 to 77% in the evening, along the coastal areas. In the Palghat region it is 79% and 63% respectively. During the period from January to march, afternoon humidity reduces to 60 to 63%, with variation from 35% in the interior to 71% in the coastal areas.

I.1.I. Flora and Fauna

The variations in climate and seasons have had their impact on vegetation and the development of agriculture. Kerala is also rich in fauna, flora and mineral resources. Its forests abound in a variety of animals and birds. Among the exports from ancient Kerala to foreign countries the elephant, the peacock and the monkey seem to have had the place of honour. Ivory was also a valuable item of export. The aromatic plants and spices of Kerala attracted the attention of foreigners even from time immemorial. Such spices as pepper, cardamom, cinnamon and ginger were exported from ancient Kerala to the countries of Asia and Europe and they continue to earn valuable foreign exchange even now. It was the demand for the pepper ('black gold') of Kerala that brought European powers like the Portuguese, the Dutch, the French and the English 'to the Kerala coast in modern times and led to the establishment of European domination over the country.

The forests of Kerala abound in some of the rarest species of timber and they have been very much in demand in the foreign markets. The teak-wood from Kerala found its way to foreign countries even centuries before the dawn of the Christian era as is evidenced by the discovery of teak in the ruins of Ur. The magnificent teak of the Kerala forests appears to have been used for the manufacture of the ships. Kerala is also famous for its rich mineral resources. Kautilya's Arthasastra (4th century B.C.) refers to the river Churni (Periyar) as one of the places in India where pearls could be found. The ancient Romans carried from Kerala such luxuries as pearls and diamonds in exchange for their gold.

Model Questions

I SEMESTER BA (CUFYUGP) EXAMINATIONS

HIS1CJ101 History of Keralam upto 12^{th} Century

Time: 2 Hours Maximum Marks: 70

Section A

(Answer all questions. Each carries **3 marks**)

(Ceiling: 24 Marks)

1. Geomorphology
2. Western Ghats
3. Mountain pass
4. Backwaters
5. Estuaries
6. Monsoon
7. Laterite soil
8. Biodiversity
9. Midland
10. High ranges

Section B

(Answer any **8** Questions. Each carries **6 Marks**)

(Ceiling: 36 Marks)

11. Explain the concept of Keralam as a distinct ecological region.
12. Describe the significance of the Western Ghats in shaping the environment of Keralam.
13. Discuss the major passes in the Western Ghats and their importance.
14. Examine the role of the Indian Ocean in influencing Kerala's climate and trade.
15. Write a note on the significance of rivers in shaping the history of Keralam.
16. Discuss different types of soil in Kerala and their distribution.
17. Explain the monsoon system and its impact on the economy of Keralam.
18. Describe the medical and historical significance of the flora and fauna of Keralam
19. Examine the ecological importance of backwaters and their role in Kerala's economy.
20. Discuss the impact of climate and monsoons on Kerala's agriculture and lifestyle.
21. Evaluate the richness of flora and fauna and their significance in the local medical system

Section C

Answer any **One** of the Following

(1 × 10 = 10 Marks)

21. Critically examine Keralam as a unique geographical and cultural region, highlighting its natural boundaries ?

22. Evaluate the ecological and economic significance of the Western Ghats in Keralam.

23. Discuss the role of mountain passes in facilitating cultural and commercial interactions in Keralam’s history.

24. Analyze the relationship between the Indian Ocean and Keralam’s historical development, especially in terms of climate and trade.

25. Provide a detailed account of the river systems of Keralam and their role in shaping settlement patterns.

26. Analyze the distribution and characteristics of soil types in Kerala and their agricultural implications.